

ANUSHKA EKKKA

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EDUCATION

International Institute of Information Technology, Naya Raipur

July 2023 - July 2027

B.Tech in Data Science and Artificial Intelligence

8.73/10

WORK EXPERIENCE

Samsung Research & Development - R&D Intern

September 2025

- Designed and implemented robust error-handling and validation pipelines for LLM outputs, addressing multiple edge-case categories (ambiguous queries, missing context, conflicting inputs) and improving structured response reliability by 25–35%

IIIT Naya Raipur - Research Assistant

Aug 2024 - Dec 2024

- Modeled a machine learning model using liver function tests for disease classification, achieving **99.94% accuracy** using L2-regularized XGBoost.

Chhattisgarh Police - Full Stack Website Developer

Jun 2024 - Present

- Developed the official Chhattisgarh Police government website under a Digital Initiative, implementing role-based access control, secure APIs, and a scalable architecture to support 5,000+ users/month with 99% uptime and improved page load time by 30%.

ACHIEVEMENTS

- Winner** - Engineer's Ring of Honour (GeeksforGeeks x Naukri.com, 1st Place, 2025) , Government of India IDEA ONE Hackathon 2025 (among 600+ teams) , Chhattisgarh Envirothon 2025 for Sustainable AI Solutions(among 100+ teams)
- Finalist in 8+ Hackathons** - Including Microsoft Code Cubicle 5.0 (3rd Runner-up)
- selected as a mentee for the Codess.Cafe**
- Solved 1000+ DSA problems** on LeetCode, **4 star** on CodeChef and **Expert** Codeforces ([LeetCode](#) — [Codeforces](#) — [CodeChef](#))
- Selected for Infosys Pragati Cohort 2 and SheFi Scholar 2025**

PUBLICATIONS

IEEE NetCrypt 2025, JNU - Detectify: Optimized Network Traffic Anomaly Detection - [Link](#)

PROJECTS

Walmart-YOLO: AI System for Real-Time Waste Management — *Python, YOLOv8, Q-Learning, Pathway CDC*

[GitHub](#)

- Architected and deployed an end-to-end AI-driven inventory management system, approved by Chhattisgarh Government / NABARD, supporting real-time monitoring of 1,500+ SKUs across multiple CG Mart outlets, reducing manual inventory intervention and enabling a scalable, closed-loop retail ecosystem.
- Automated price adjustments for 50+ perishable items across 8 inventory states using Q-learning, decreasing food spoilage by 25% and boosting near-expiry stock sell-through rates.
- Integrated YOLOv8 for real-time visual tracking and automated inventory counting, fusing computer vision with master product data via Pathway-based low-latency CDC pipeline ensuring system integrity and horizontal scalability across thousands of SKUs.

CGPD-OCR: DevanagariDeepScan — *Python, PyTorch, CRNN-CTC, OpenCV, Offline LLM*

[GitHub](#)

- Built a **secure offline AI system** for Chhattisgarh Police combining handwritten Hindi OCR and a station-level LLM chatbot, enabling real-time queries across **1,000+ digitized records** with **100% on-prem data privacy**.
- Developed a **CRNN-CTC OCR pipeline** trained on **5,000+ word images**, achieving **92–95% accuracy** and delivering **3× performance improvement** on noisy, stamped police documents.
- Applied **multi-source training and retrieval** to handle high handwriting variance, enabling **instant keyword search and case analysis** and reducing manual lookup time from **hours to seconds**.

Bidirectional Neural Transliteration Service (Kaithi Hindi)— *Python, GANs, Deep Learning, Frequency Domain Analysis*

[GitHub](#)

- Engineered a custom Sequence-to-Sequence (Seq2Seq) LSTM neural network in PyTorch to automate character-level transliteration, achieving **94% validation accuracy**.
- Architected a highly scalable REST backend with FastAPI to serve the PyTorch inference engine, exposing 2 core bidirectional endpoints that process real-time text queries with **50ms latency** and can seamlessly handle 100+ concurrent requests per second.
- Compressed the trained model footprint to just 900 KB, reducing memory overhead by **95%** compared to standard NLP models and enabling highly efficient inference.

TECHNICAL SKILLS

Technical Skills & Tools: Python, C++, Java, C, SQL, JavaScript, HTML/CSS, VS Code, Git, GitHub, Jupyter Notebook, Flask, MySQL, Jenkins, Eclipse, Google Cloud Platform, Android Studio, WordPress

Core Skills: Machine Learning, Deep Learning, Reinforcement Learning (Q-Learning), Computer Vision (YOLOv8, OpenCV), Natural Language Processing, Explainable AI (XAI, SHAP, LIME), XGBoost, TensorFlow, PyTorch, GANs, CRNN-CTC, Transformers, IoT, Data Pre-Processing, Feature Engineering, JSON Schema Validation

Soft Skills: Problem Solving, Leadership, Communication, Public Speaking, Event Execution and Management, Team Coordination

CLUBS & RESPONSIBILITIES

Head Placement Coordinator - Training and Placement Cell, IIIT-NR

2024 - Present

Science and Technology Secretary, IIIT-NR

2023 - 2024

Documentation Coordinator - MUN Club, IIIT-NR

2023 - 2024